Atrey Desai

Brendan Iribe Center for Computer Science & Engineering
University of Maryland
College Park, MD 20742

adesai10@umd.edu
atreydesai.com
Google Scholar

Education

University of Maryland, College Park *Exp. Graduation: May 2027*
B.S. in Computer Science (Honors) & B.A. in Linguistics (Minor in Korean Studies) * – in progress
Advisors: Prof. Rachel Rudinger & Prof. Jordan Boyd-Graber
Coursework: Natural Language Processing (Grad), Commonsense Reasoning (Grad), Mechanistic Interpretability (Grad)*, Machine Learning, Data Science, Parallel Computing, Computer Systems, Syntax, Semantics*, Phonetics, Psycholinguistics

Experience

University of Maryland, College Park *May 2024 – present*
Undergrad Researcher, CLIP Lab
Advisors: Prof. Rachel Rudinger, Prof. Jordan Boyd-Graber, & Prof. Mohit Iyyer

- Quantified LLM susceptibility to human-written vs. LLM-generated MCQs to see if questions contain unintended artifacts and are solvable without full context; created tooling to isolate distractor-choice provenance effects and analyzed the reliability of human synthetic data for model evaluation and distillation.
- Constructed an adversarial benchmark to test VLM capabilities in detecting out-of-context (OOC) video-based misinformation on social media based on multimodal clues and user interactions to probe reasoning under deceptive framing.
- Applied causal interpretability techniques to demonstrate that LMs require significantly more data than humans to acquire general syntactic representations, revealing heightened sensitivity to item-level and construction-level variation.

The University of Texas at Arlington *Feb 2024 – present*
Visiting Researcher, ACL2 Lab & NSF
Advisor: Prof. Kenny Zhu

- Constructed AniVoice-cat, a dataset of 26,000+ annotated cat vocalizations from 250+ hours of video; identified 57 unique cat phones, creating a foundation for future computational research on non-human communication.
- Improved transcription pipeline using PANNs and HuBERT models to 96% accuracy with 93% top-5 accuracy in action recognition, setting a new state-of-the-art for automated animal vocalization analysis.

Learn Prompting *May 2025 – Jan 2026*
Member of Technical Staff (San Francisco, CA)

- Oversaw Hackaprompt, the world's largest red-teaming hackathon, managing user comms and resolving technical issues.
- Researched the robustness of AI safety judges against adversarial CBRNE-related universal and multimodal jailbreaks.

University of Maryland, College Park *Dec 2023 – Aug 2024*
Researcher, FIRE Sustainability Analytics Lab
Advisor: Prof. Thanicha Ruangmas

- Developed an automated OCR pipeline for environmental impact assessments of U.S. emissions based on 50,000+ 10-K filings and crafted a framework to guide evidence-based policymaking on climate restoration strategies.

Brown University *Dec 2020 – June 2023*
Researcher, Reinforcement Learning at Brown Group
Advisor: Prof. Michael Littman

- Developed a custom RL environment showing non-experts can solve complex tasks via reward functions and agent behavior; found human-readable interfaces give fine-grained inference control and ease human-AI interaction in robotics.

Publications

Preprints

* – equal contribution

2. Quick, Create a Distractor! Evaluating LLM Distractors for Multiple-Choice Benchmarks
   Under Review (ACL ARR)
   Atrey Desai, Nishant Balepur, Rachel Rudinger

1. A Preview of Computational Animal Linguistics
   Under Review (ACM Computing Surveys)
   Tuan Dang*, **Atrey Desai***, Hridayesh Lekhak*, Tirza Panunto*, Lindsay Pike*, Theron Wang*, and Kenny Zhu*

Publications

5. Filling in the Mechanisms: How do LMs Learn Filler-Gap Dependencies under Developmental Constraints?
   ACL 2026 (Findings), TLS 2026 (Oral)
   Atrey Desai and Sathvik Nair

4. BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks
   ACL 2026
   Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, **Atrey Desai**, . . . , Jordan Boyd-Graber
3. Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers
   ACL 2026
   Nishant Balepur, **Atrey Desai** and Rachel Rudinger
2. Language Models Generate Multiple-Choice Questions with Artifacts
   MASC-SLL 2025
   Atrey Desai, Nishant Balepur and Rachel Rudinger
1. Reinforcement Learning As End-User Trigger-Action Programming
   IML Workshop @ AAAI 2022; RLDM 2022
   Chace Hayhurst, Hyojae Park, **Atrey Desai**, Suheidy De Los Santos and Michael Littman

Selected Awards and Honors

- **CRA NSF CISE Student Funding Program** ($9,000) *2026*
- **Department of Computer Science Professional Development Fund** ($500) *2026*
- **SPIRE Research Grant** ($3,000) *2025*
- **Omicron Delta Kappa Top 10 Freshman** *2024*
- **CMSC & ARHU Dean's List** *2023 – 2026*
- **UMD President's Scholarship** ($50,000) *2023*
- **NMSC National Merit Scholarship** ($4,000) *2023*
- **Catherine Yang Scholarship** ($1,000) *2023*

Talks

- **Filler-Gap Dependencies under Developmental Constraints in LMs**
  Texas Linguistics Society; UMD Comp. Cognitive Science Reading Group *Feb 2026*
- **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
  University of Maryland; University of Texas at Arlington *Nov 2024, Jul 2024*

Professional Responsibilities

- **Organizer**, QANTA/EMM-QA (2026) *2026 – present*
- **Review**, *ACL/ACL Rolling Review (ARR), ICML EMM-QA *2025 – present*
- **Boardmember**, University Research Advisory Board *2025 – present*
- **Vice Chair**, Computer Science Department Council *2025 – 2026*
  Appointed by faculty and student body; represent 4,200+ CS undergraduates.
- **Senior Member**, FIRE Student Leadership Council *2024 – 2026*
  Represented 1000+ peers, ran events & workshops, and reformed class curricula.
- **University Ambassador (Department & College)** *2024 – present*
- **Mentorship:** 7 undergrads via UMD Office of Undergraduate Research (2025) & Technica (2024); designed and led robotics curriculum for MSET (2020–2022).

Skills

- **Languages:** Python, Java, JavaScript/HTML/CSS, R, MATLAB.
- **Libraries/Frameworks:** Huggingface (Datasets, Transformers), NLTK, PyTorch, Selenium, BeautifulSoup.
- **Tools:** Git, Docker, GCP, Google Vertex AI, VS Code.
- **Natural Languages:** English (Native), Gujarati (Native), Spanish (Intermediate), Korean (Beginner).